Power Budget

Team Number:	308
Project Name:	**Underwater Explorer**
Team Member Names:	**Sam B, Sam M, Jacob D., Andrew, Adrian,**
Version:	**1**

A. List ALL major components (active devices, integrated circuits, etc.) except for power sources, voltage regulators, resistors, capacitors, or passive elements							
All Major Components	**Component Name**	**Part Number**	**SupplyVoltageRange**	**#**	**AbsoluteMaximumCurrent (mA)**	**TotalCurrent(mA)**	**Unit**
	ESP32	ESP32-S3-WROOM	3.3V	1	800	800	mA
	OLED	NHD-2.7-12864WDW	3.3V	1	100	100	mA
	Joystick	THB001P	3.3V	2	1	2	mA
	3.3V regulator	AP63203WU-7	3.3V		2000	2000	mA
B. Assign each major component above to ONE power rail below. Try to minimize the number of different power rails in the design.							
+3.3V Power Rail	**Component Name**	**Part Number**	**SupplyVoltageRange**	**#**	**AbsoluteMaximumCurrent (mA)**	**TotalCurrent(mA)**	**Unit**
	ESP32	ESP32-S3-WROOM	3.3V	1	800	800	mA
	OLED	NHD-2.7-12864WDW	3.3V	1	100	100	mA
	Joystick	THB001P	3.3V	2	1	2	mA
					Subtotal	902	mA
					Safety Margin	25%	
					Total Current Required on 3.3V Rail	1127.5	mA
c2. Regulator or Source Ch	3.3V regulator	AP63203WU-7	3.3V		2000	2000	mA
					Total Remaining Current Available on +3.3V Rail	872.5	mA
C. Power Supply							
External Power Source 1	**Component Name**	**Part Number**	**SupplyVoltageRange**	**Output**	**AbsoluteMaximumCurrent (mA)**	**TotalCurrent(mA)**	**Unit**
Power Source 1 Selection	Plug-in Wall Supply	C/DC ADAPTOR 093	100-240VAC	+9V	3000	3000	mA
Power Source 2 Selection	USB Port		5V	5V	2000	2000	
Power Rails Connected to External Power Source 1							
	3.3V regulator	AP63203WU-7	3.3V		2000	2000	mA
					Total Remaining Current Available on External Power Source 1	1000	mA
Power Rails Connected to External Power Source 2							
	3.3V regulator	AP63203WU-7	3.3V		2000	2000	mA
					Total Remaining Current Aailable on External Power Source 2		
						0	mA